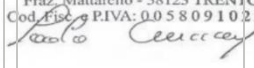


UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIS 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

IDENTIFICATION TABLE

Client	PBDiM - Energopol
Project	Udrożnienie Łódzkiego Węzła Kolejowego (TEN-T), Etap II, Odcinek Łódź Fabryczna - Łódź Kaliska/Łódź Żabieniec – Projekt POIIS 5.1-15
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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

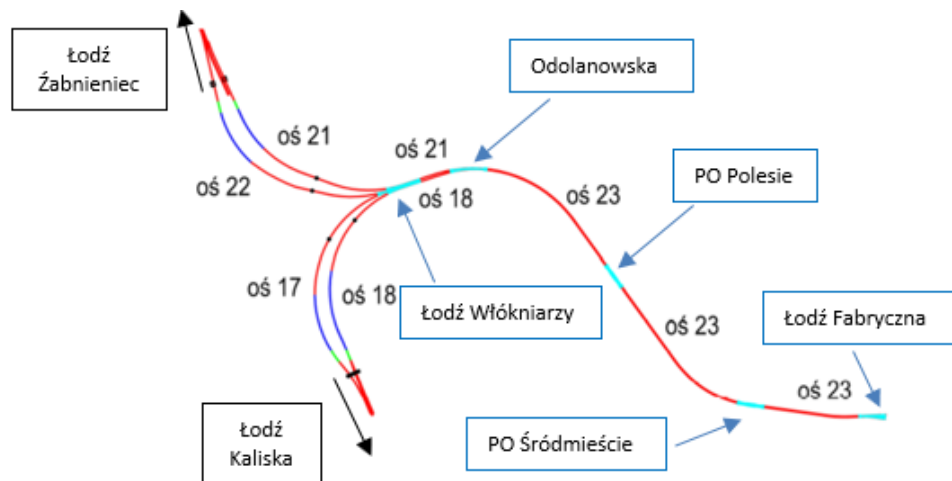


Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 09/09/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+628.92 after the erection of ring n. 269; which corresponds to a total excavation length of 337.55 m from the starting chainage.

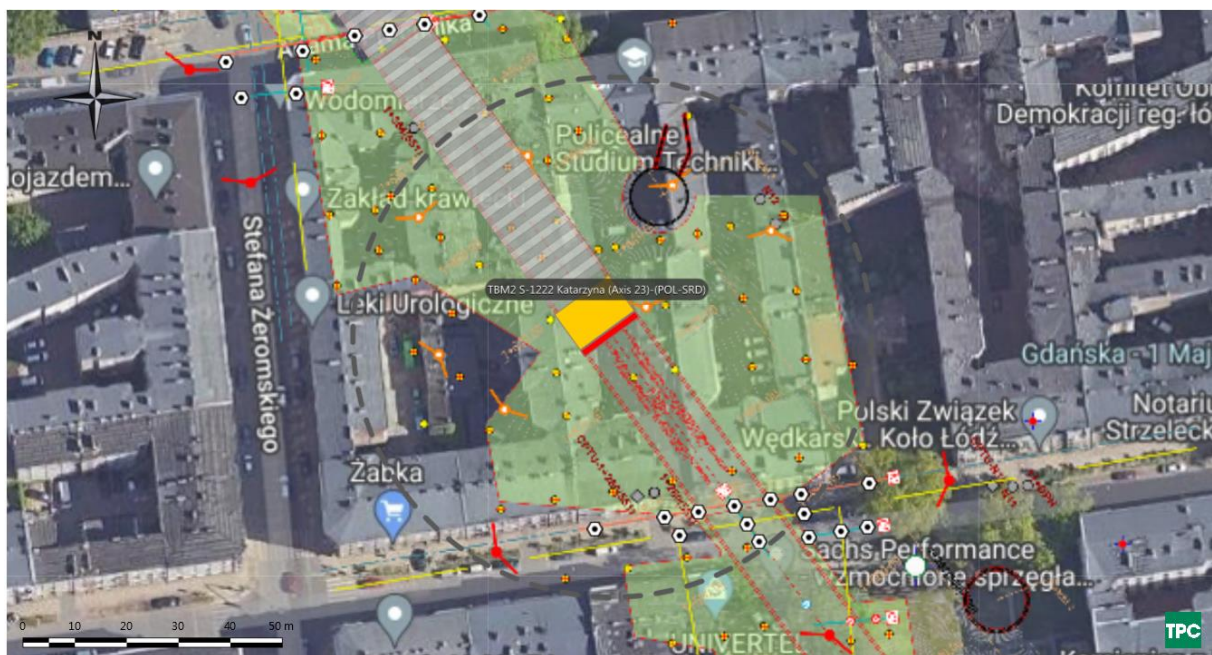


Figure 1-2 Plan view of TBM Position on 10/09/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.8 bar. The value is within the attention values. Figure 2-1 shows EPB sensors.

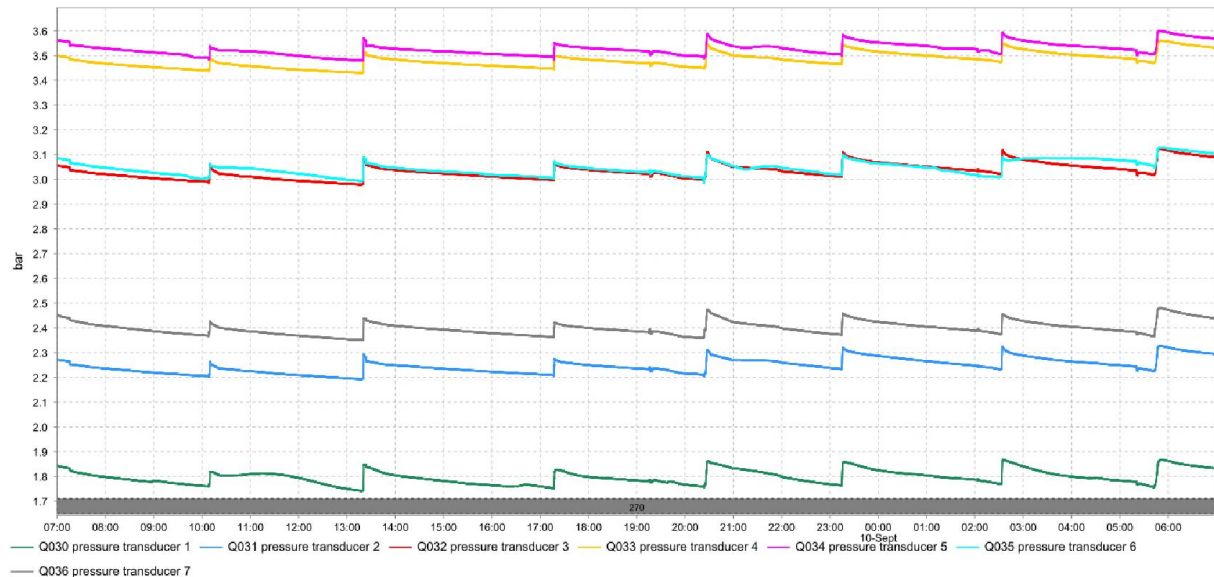


Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.45 bar indicating that the bulk chamber is filled.

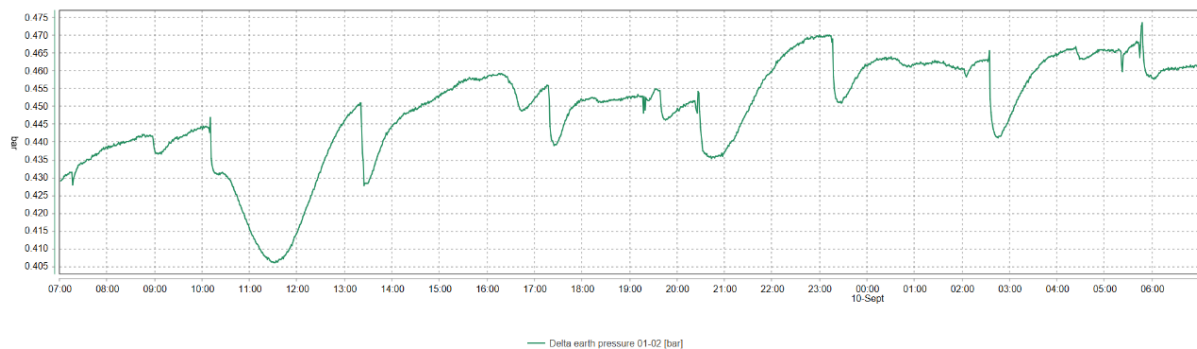


Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values lower than n. 1. An increase in weight was observed during the stoppage, attributed to the cleaning process and the pumping of grout from the tank. The belt scale value is a raw indication of the excavated material quantity.

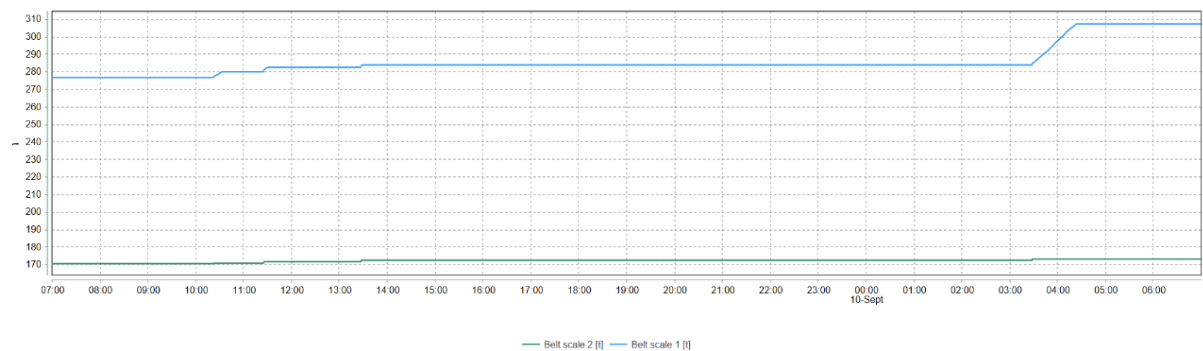


Figure 2-3 Belt scales

3. THRUST AND CUTTERHEAD TORQUE

Graph for torque, advance rate and thrust force are below shown. The torque graph shows non-zero values due to the rotation of the cutterhead.

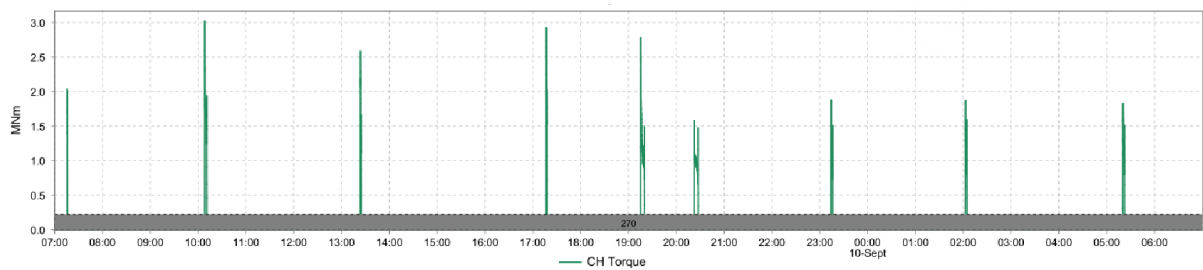


Figure 3-1 CH Torque

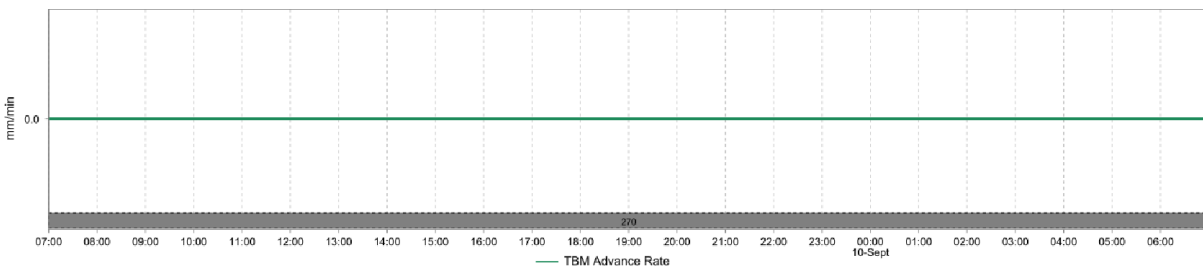


Figure 3-2 Advance rate

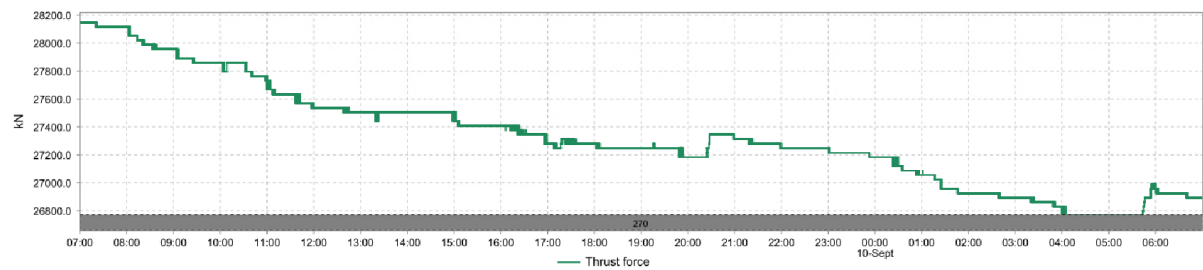


Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection volume has to be compared to minimum theoretical annular volume $\pm 10\%$ of 13.54 m^3 at the end of ring excavation. Ratio B/A is at 5%.

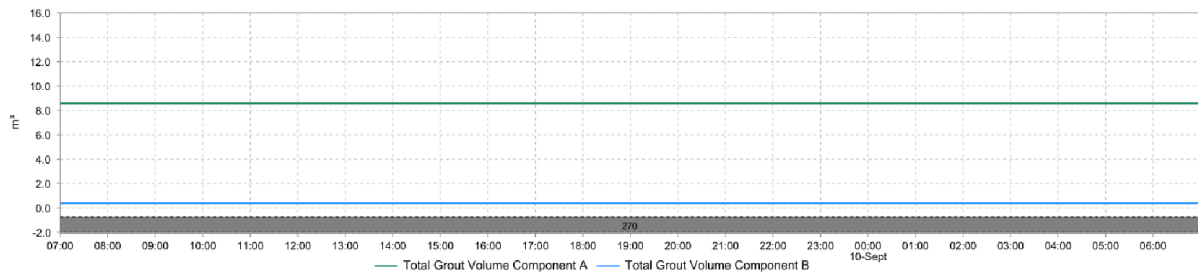


Figure 4-1 Grout volumes

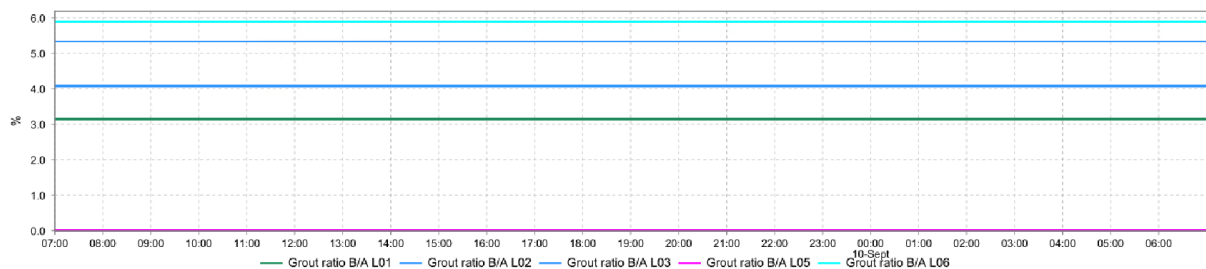


Figure 4-2 Volume ratio Components B/A

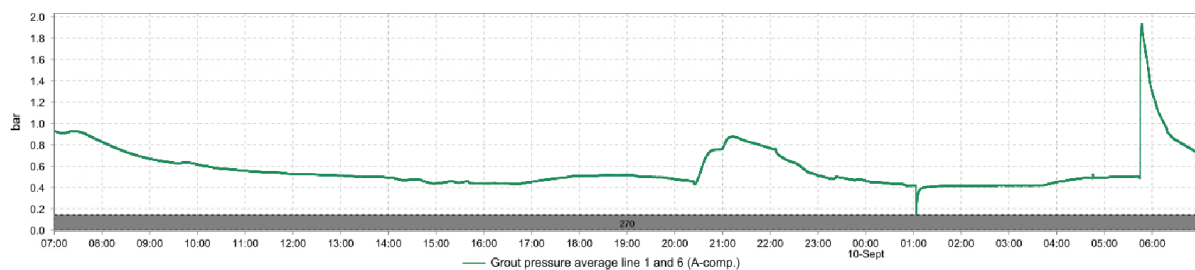


Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about -15.85 mm on the front edge of the shield and the horizontal deviation of about -17.86 mm (Figure 5-1). Maximum radial deviation is of 24.71 mm.

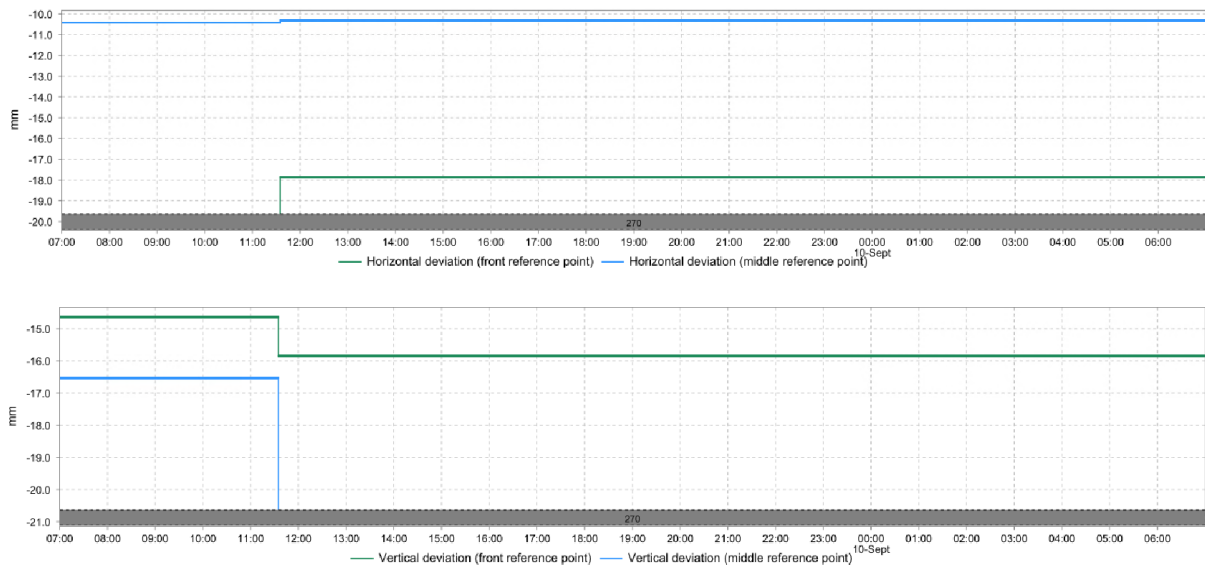


Figure 5-1 Machine deviation from the alignment

6. MONITORING

Based on data from Geoscope and Beyond, building subsidence has been observed but is showing signs of stabilization.

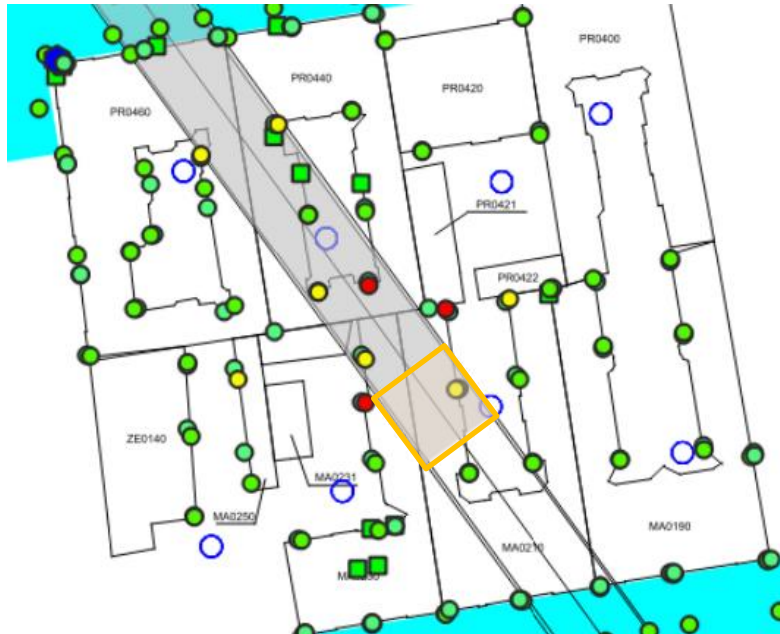


Figure 6-1 Monitoring points – Geoscope

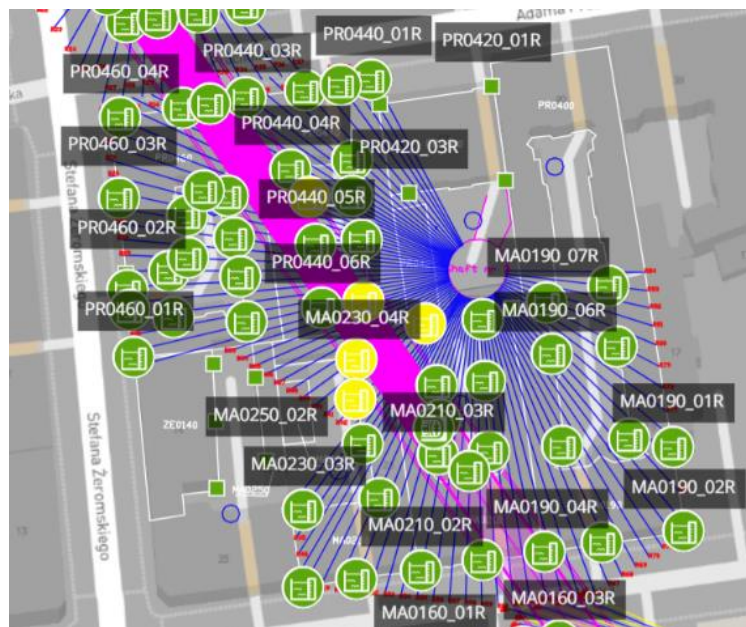


Figure 6-2 Monitoring points – Beyond

On PR0440, Geoscope monitoring points show stabilizing trends starting from 07/09/2024 at 8:00 am, following the excavation stoppage. It is not possible to appreciate a complete stabilization trend because many points have not recorded new measurements since 08/09/2024 in the morning.

In Beyond, monitoring points show stabilizing trends, except for PR0440_06_20L and PR0440_07_20L which showed an uplift followed by a new downward trend.

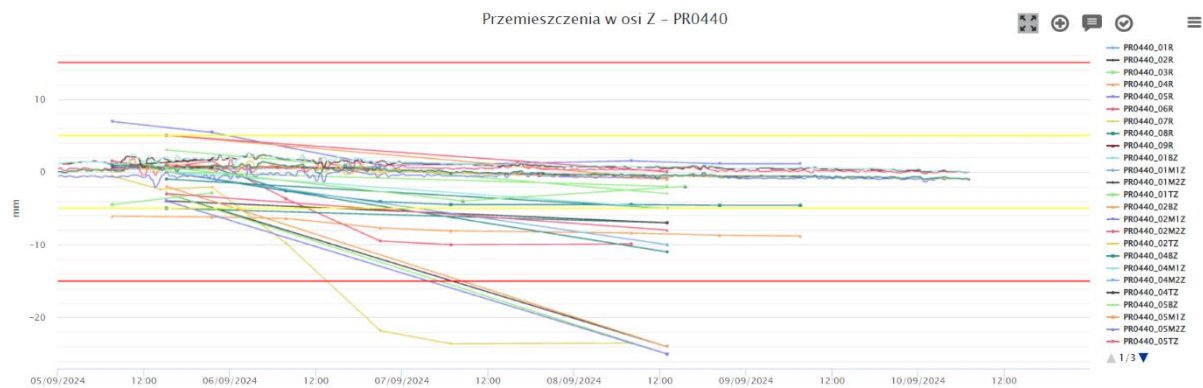


Figure 6-3 Settlement of building PR0440 - Geoscope



Figure 6-4 Monitoring point PR0440 - Beyond

On PR0460, Geoscope monitoring points show a small downward trends for point PR0460_06. Other points show stabilizing trends, included the PR0460_07 point that has exceeded the attention threshold. Point PR0460_09 shows a small uplift trend.

In Beyond, monitoring points show stabilizing trends, except for PR0460_09_18L and PR0460_16_18L which showed an uplift followed by a stabilizing downward trend.

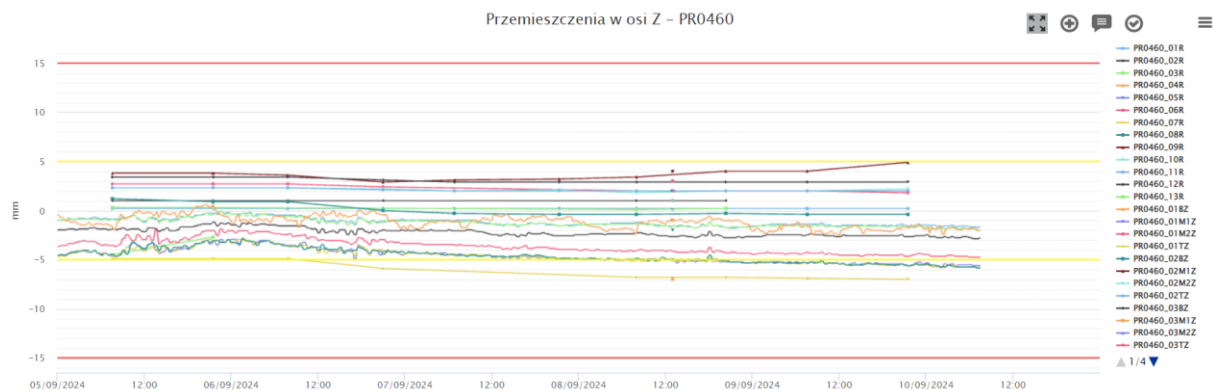


Figure 6-5 Settlement of building PR0460 - Geoscope



Figure 6-6 Monitoring point PR0460 - Beyond

On MA0190, monitoring points are stable.

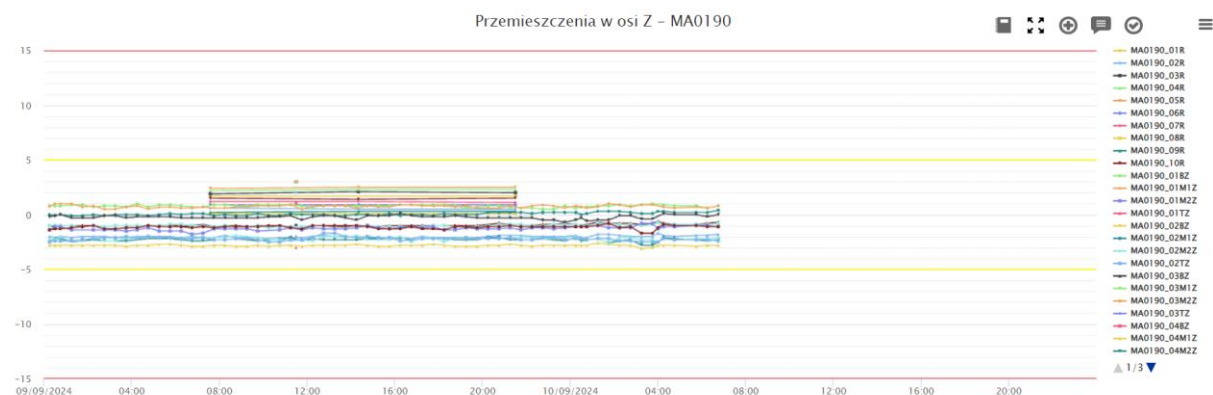


Figure 6-7 Settlement of building MA0190 - Geoscope



Figure 6-8 Monitoring point MA0190 - Beyond

On MA0210, Geoscope monitoring points show stabilizing trends starting from 07/09/2024, following the excavation stoppage. It is not possible to appreciate a complete stabilization trend because many points have not recorded new measurements since 08/09/2024 in the morning.

In Beyond, monitoring points show stabilizing trends, except for MA0210_05_22L which showed an uplift followed by a new downward trend.

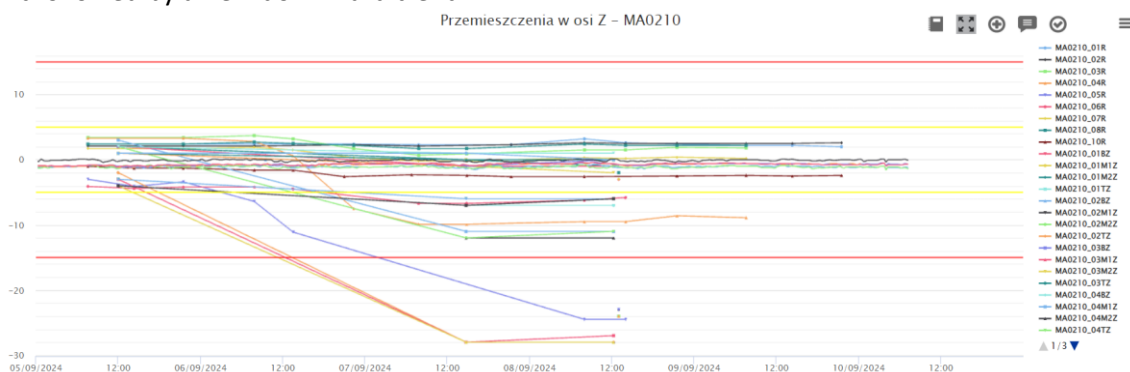


Figure 6-9 Settlement of building MA0210 - Geoscope



Figure 6-10 Monitoring point MA0210 - Beyond

On MA0230, Geoscope monitoring points show stabilizing trends starting from 07/09/2024, following the excavation stoppage. It is not possible to appreciate a complete stabilization trend because many points have not recorded new measurements since 08/09/2024 in the morning.

In Beyond, monitoring points show stabilizing trends.

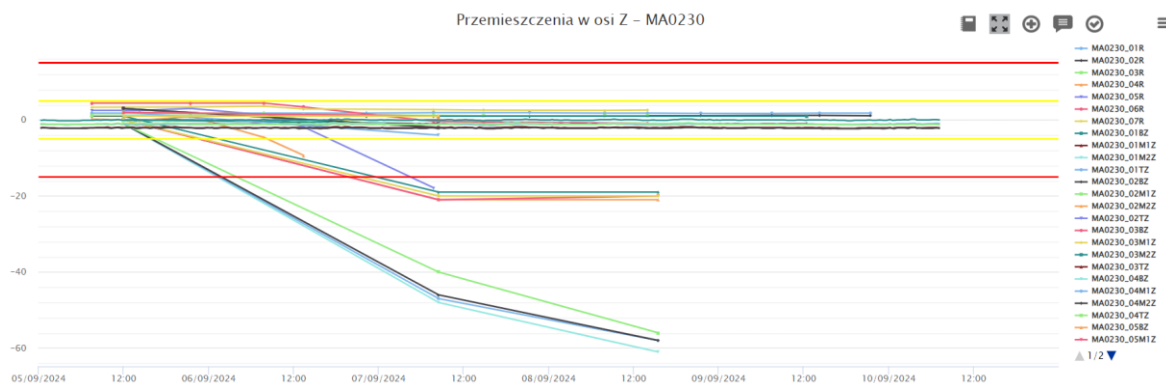


Figure 6-11 Settlement of building MA0230 - Geoscope



Figure 6-12 Monitoring point MA0230 - Beyond

On MA0250, monitoring points show an uplift trend, exceeding the attention threshold.

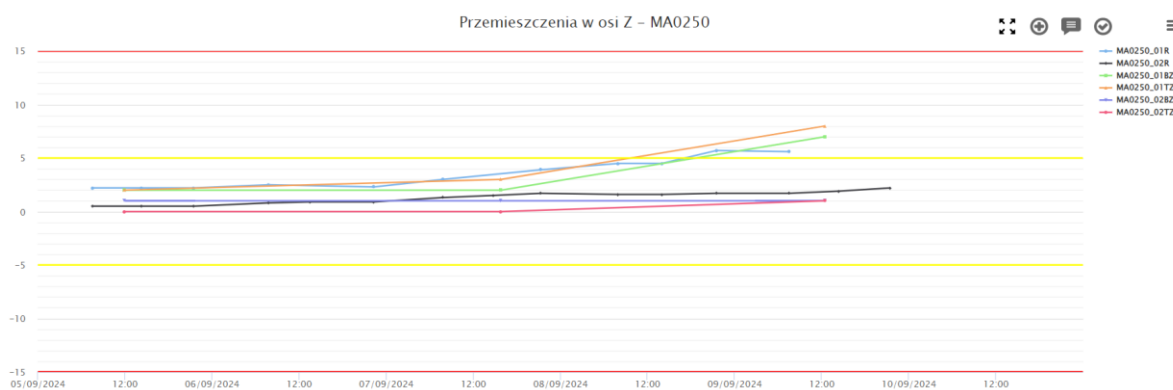


Figure 6-13 Settlement of building MA0250 - Geoscope

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 09/09/2024 of Soletanche, compensation grouting has been performed on the 09/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 15258.63 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure.

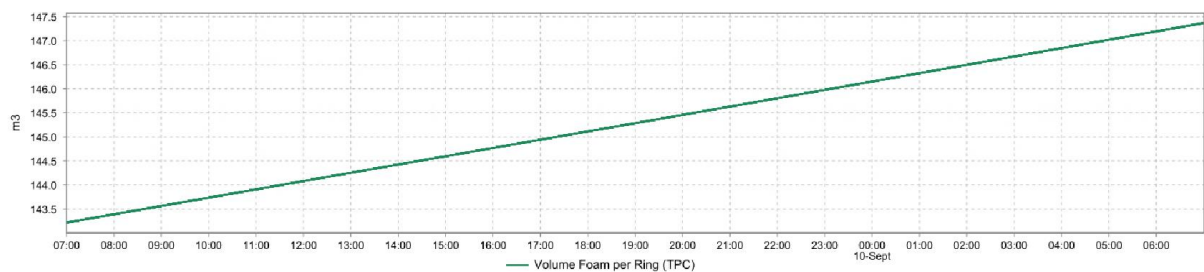


Figure 8-1 Injected foam volume

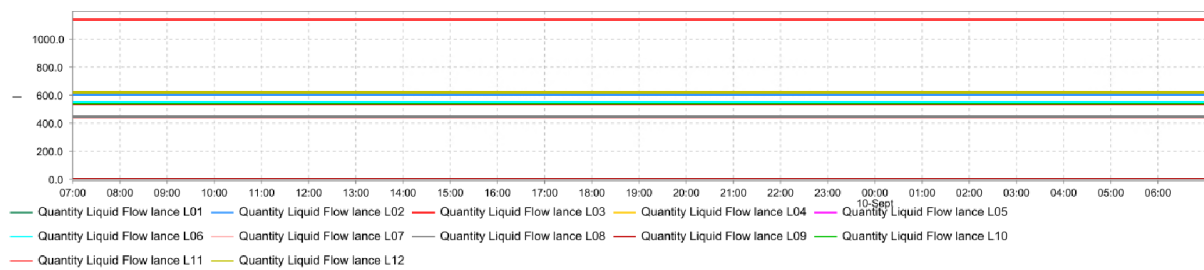


Figure 8-2 Conditioning liquid volume